Amazon Product Dataset Analysis: Executive Summary

Role: AI-First Analytics Engineer Assessment | **Candidate:** anujhsrsaini | **Date:** October 2026
Live AI Assistant: Antigravity CLI (Google Gemini 3.8 Flash) | **Repo:** github.com/anujhsrsaini/Pearl-Assignment

Context & Approach: *This evaluation simulates how an AI-first analyst operates under live constraints (30 minutes) on unfamiliar, scraped eCommerce data (~1,000 Amazon records). Rather than manual ad-hoc scripts, autonomous specialized AI agents were orchestrated to inspect data syntax, profile schema distributions, and clean anomalies before answering strategic business questions.*

1. Data Orientation: What is this data, and would you trust it?

The Reality: The dataset contains 1,000 scraped Amazon product records across 55 columns. In its raw form, **I would not trust this data for any commercial business decision.**

- **Silent Parsing Failures:** A literal newline (\n) in Row 320's discount field split the record across lines 321 and 322, corrupting line-by-line parsers into 27- and 29-column fragments. Extreme JSON arrays in variation rows (up to 151 KB) trigger buffer crashes in standard Python CSV modules.
- **Price Corruption:** Over 85% of final_price values were escaped with triple quotes (""57.79""), and 177 prices were in scientific notation (1.795e+01). Direct numeric casting fails immediately without regex sanitization.
- **Ghost Columns & Redactions:** Three columns (product_details, prices_breakdown, country_of_origin) are 100% empty. Furthermore, 801 seller names were obfuscated with black blocks (■).
- **What was fixed first:** Before conducting analysis, an automated cleaning pipeline dropped dead columns, sanitized prices to floats, recovered seller identities using unmasked buybox data, stripped invisible BiDi marks (\u200e/\u200f), and expanded JSON category trees into 7 queryable columns.

2. Business Question: "Which product category should we prioritise, and why?"

Executive Recommendation to Category Manager: Prioritise **Home & Kitchen** as the primary growth pillar, paired with **Tools & Home Improvement** for rapid market entry.

Category	Catalog Share	Total Reviews	Median Reviews	Mean Rating	Strategic Viability Assessment
Home & Kitchen	15.0% (150)	417,487 (12.9%)	498.5	4.47 / 5.0	**Primary Recommendation:** High customer delight (4.47), large addressable catalog, manageable review barrier (~500 reviews).
Tools & Home Improvement	11.2% (112)	104,117 (3.2%)	221.5	4.38 / 5.0	**Low-Hanging Fruit:** Lowest review barrier among major categories (221 median). Fast path for new listings to achieve search parity.
Clothing, Shoes & Jewelry	21.6% (216)	951,495 (29.3%)	924.5	4.36 / 5.0	**Saturated / High Moat:** High volume, but incumbent review moat (>920 median) requires massive ad spend to compete.
Pet Supplies	2.6% (26)	409,036 (12.6%)	2,007.5	4.35 / 5.0	**Extreme Review Moat:** Incredible review density (2,008 median reviews), but entry barriers for unproven brands are severe.
Beauty	3.1% (31)	110,555 (3.4%)	114.0	4.06 / 5.0	**High Risk:** Lowest customer rating on Amazon (4.06 vs 4.40 platform avg). Elevated return and dissatisfaction rates.

- **What the data supports:** Catalog presence, relative social proof intensity (review counts), and customer sentiment distributions (ratings) across categories.

- **What the data DOES NOT support:** Revenue, unit velocity, profit margins, conversion rates, or return rates. We can measure listing dominance, not commercial profitability.

3. Data Skepticism: What you must verify before taking action

- **Phantom Discounts (87 Rows):** In 87 listings, initial_price == final_price down to the cent, yet non-zero discounts (up to -82%) were recorded. Raw scrape discounts are marketing badges, not financial truths.
- **Multi-Currency Poisoning:** The dataset mixes USD (96.7%), INR (3.1%), and GBP (0.2%). Treating final prices as homogeneous numbers without FX normalization produces completely distorted price brackets.
- **Unverified Velocity (bought_past_month):** 79.6% null, and the remaining values are coarse Amazon rounded estimates (e.g. 50, 100, 1000). It cannot be used for demand forecasting.

4. Something You Didn't Ask: Observations Beyond the Brief

- **Security / Scraping Leak:** While seller_name was intentionally redacted with block glyphs (■) in 801 rows for anonymity, the scraper inadvertently left buybox_seller completely unmasked. This allowed us to 100% reverse-engineer vendor identities and map Amazon 1P vs 3P buybox control.
- **The Question I Wanted to Answer but Couldn't:** *"Does Amazon prioritize its own private-label products (Amazon Essentials) over 3P merchants in the Buy Box when competing on identical price?"* — Cannot be definitively proven because competitor price bid logs (prices_breakdown) are 100% dead.

5. AI Workflow & Conversation Links

Every step of this analysis — including prompt chains, agent tool execution, raw transcripts, and code — is permanently committed to GitHub:

- **Full Session Transcripts:** https://github.com/anujhsrsaini/Pearl-Assignment/tree/main/antigravity_conversations
- **Reproducible Data Pipeline:** https://github.com/anujhsrsaini/Pearl-Assignment/blob/main/src/clean_amazon_products.py
- **Detailed Audit Report:** https://github.com/anujhsrsaini/Pearl-Assignment/blob/main/reports/amazon_products_audit_report.md